# L05: Fundamentals of Machine Learning

## CSci 560 Deep Learning w/ Python (Chollet) Ch. 5

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# L05.1 Generalization: The goal of machine learning

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# Generalization: The goal of machine learning

**Fundamental Issue of ML**

The fundamental issue in machine learning is the tension between optimization and generalization.

- After just a few epochs of training, performance on never-before-seen data starts to diverge
  - The models start to **overfit**
- **Optimization** refers to the process of adjusting a model to get the best performance possible on the training data.
- **Generalization** refers to how well the trained model performs on data it has never seen before.

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# Underfitting and Overfitting

- **underfit** the network hasn't yet modeled all relevant patterns in the training data.
- Eventually validation loss begins to increase while training loss continues to imporve
  - Generalization has stopped improving, validation metrics stall and then degrade
- The model is **overfitting**
  - It is learning patterns specific to the training data (noise)

Overfitting is particularly likely to occur when your data is noisy, if it involves uncertainty, or it if includes rare features.

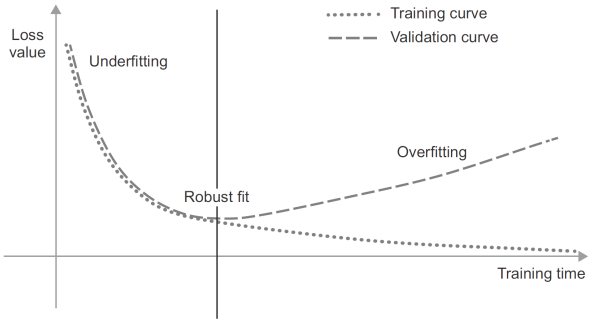


Figure 1: Canonical overfitting behavior. (Chollet, 2021, pg.122)

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# Noisy Training Data

- In real-world datasets, inputs are often **ambiguous** or **invalid**.
- Even worse, some of the data will be **mislabeled**.
- If a model goes out of its way to incorporate such outliers, its generalization performance will degrade.

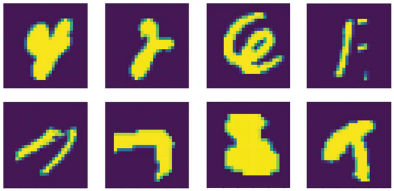

Figure 2: Weird MNIST training samples, what are these? (Chollet, 2021, pg.123)


Figure 3: Mislabeled MNIST training samples? (Chollet, 2021, pg.123)

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# Dealing with Outliers: Robust fit vs. Overfitting

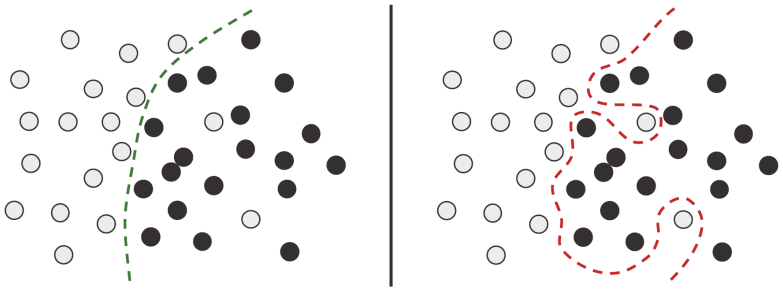

Figure 4: Dealing with outliers: robust fit vs. overfitting. (Chollet, 2021, pg.124)

- If a model goes out of its way to incorporate such outliers (ambiguous, mislabeled, noisy), its generalization performance will degrade.
- On the right, a model that overfits goes out of its way to incorporate examples that may be mislabeled.
- On the left, this model probably will generalize much better by ignoring outliers.

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# Ambiguous Features

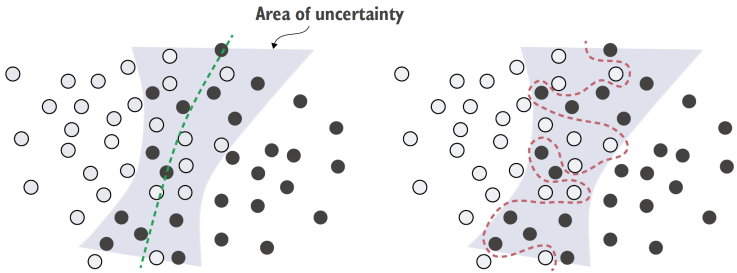


Figure 5: Robust fit vs. overfitting giving an ambiguous area of the feature space. (Chollet, 2021, pg.124)

- Not all data noise comes from inaccuracies
  - Even clean and correctly labeled data can be noisy.
- For example, the area of uncertainity might represent noise where it is difficult to differentiate between the two classes.
- A model that overfits on such noisy data will be too confident about ambiguous regions of the feature space.

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# Rare Features and Spurious Correlations

- Machine learning models trained on datasets that include rare features are highly susceptible to overfitting.
- Spurious correlations are present even in noise.
- Example training MNIST and adding noise vs. just 0's.

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# The Nature of Generalization in Deep Learning

- A deep learning model can be trained to fit anything, as long as it has enough representational power.
- You can get training loss on completely random noise. (see example)
- How can deep learning models generalize?
  - When learning noise like this example, are basically learning an ad hoc mapping between training inputs and targets, like a fancy `dict`

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# The Manifold Hypothesis

**Manifold Hypothesis**

The **Manifold Hypothesis** posits that all natural data lies on a low-dimensional manifold within the high-dimensional space where it is encoded.

- Take MNIST dataset, total possible inputs is $256^{28\times 28}$
- Very few of these inputs would look like a digit.
  - actual digits occupy a tiny *subspace* of the possible space
  - not a random subspace, it is **continuous**
  - all samples in the subspace are *connected* by smooth paths
- Handwritten digits form a **manifold** within the space of possible inputs.

The manifold hypothesis implies that:

- Machine learning models only have to fit relatively simple, low-dimensional highly structured supspaces within their potential input space.
- It's always possible to **interpolate** between two inputs.

**The ability to interpolate between samples is the key to understanding generalization in deep learning.**

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# Interpolation as a Source of Generalization

- You can make sense of the **totality** of the space using only a **sample** of the space.
- Interpolation along a manifold is different from linear interpolation in the parent space.
- Interpolation can only help you make sens of things that are close to what you've seen before.

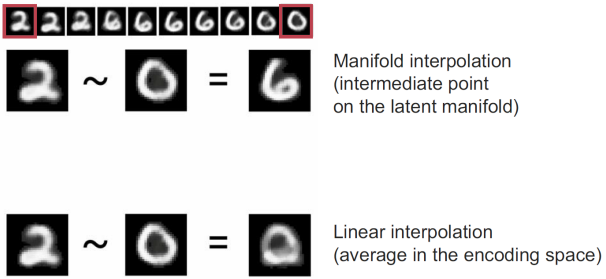


Figure 6: Difference between linear interpolation and interpolation on the latent manifold. Every point on the latent manifold of digits is a valid digit, but the average ot two digits usually isn't. (Chollet, 2021, pg.124)

# Why Deep Learning Works

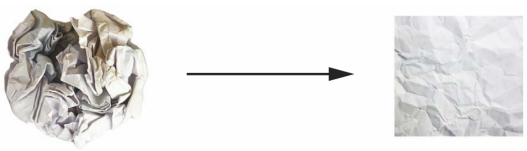

Figure 7: Uncrumpling a complicated manifold of data. (Chollet, 2021, pg.130)

- A deep learning model is a tool for uncrumpling paper balls, that is for disentangling latent manifolds.
- A deep learning model is basically a very high-dimensional curve
  - smooth and continuous (since differentiable)
  - curve fitted to data via gradient descent (smoothly and incrementally)

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# Why Deep Learning Works

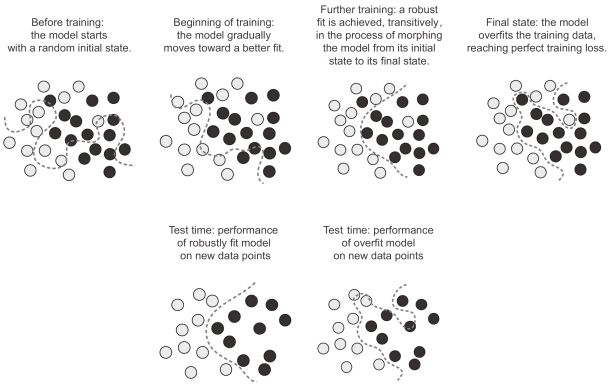


Figure 8: Going from a random model to an overfit model, and achieving a robust fit as an intermediate state. (Chollet, 2021, pg.131)

- The curve involves enough parameters that it could fit anything (even noise)
- **At some intermediate point during training the model roughly approximates the natural manifold of the data.**
- Deep learning models implement a smooth, continuous mapping from inputs to outputs. (differentiable)
- Deep learning models structured in a way that mirrors *shape* of the information in training data.

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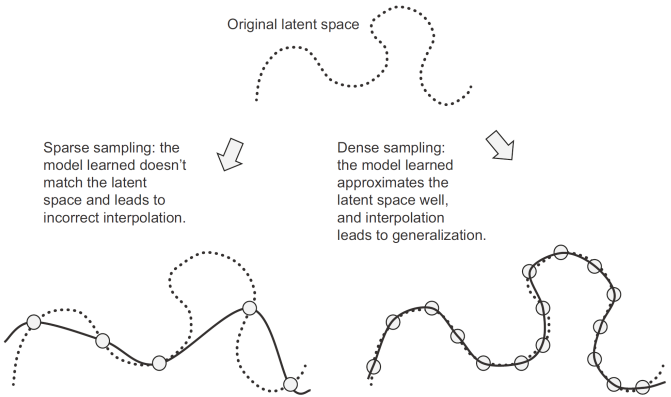
# Training Data is Paramount




- The power to generalize is a consequence of the natural structure of your data.
- Data curation and feature engineering are essential.
- For a model to perform well **it needs to be trained on a dense sampling of its input space**.
- The best way to improve a deep learning model is to train it on more data or better data.

Figure 9: A dense sampling of the input space is necessary in order to learn a model capable of accurate generalization. (Chollet, 2021, pg.131)

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# L05.2 Evaluating Machine Learning Models

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# Training Validation and Test sets

**Train / Validate / Test**

You can only control what you can observe

- It's essential to be able to reliably measure the generalization power of your model.
- You of course need some **training data** to train the model on, this is supervised learning.
- Developing a model always involves tuning its configuration.
- You need a feedback signal of the model performance while training: **validation data**
- Can **overfit to the validation set**, thus need a final **test set**

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# Simple Holdout Validation

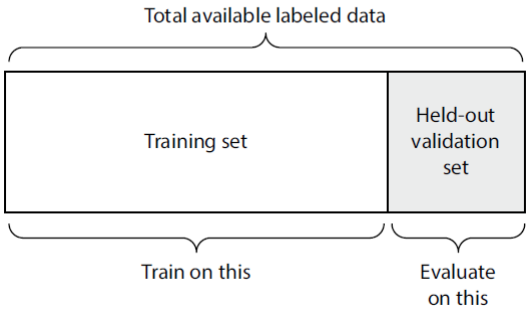


Figure 10: Simple holdout validation split. (Chollet, 2021, pg.134)

- Simple holdout validation is what we have mostly been using so far
- You can ask `keras` fit method to hold out some random part of the test data for validaiton for you.
- In order to prevent information leaks, you shouldn't tune your model based on the test set.

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# K-Fold Cross-Validation

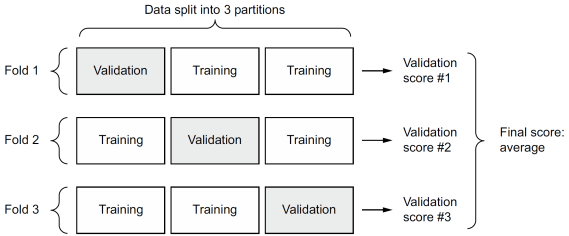


Figure 11: K-fold cross-validation with K=3 folds (Chollet, 2021, pg.135)

- If little data is available then your validation and test sets may contain too few samples to be statistically representative of the data.
- If different random shuffling rounds of data yield very different measures of performance, then you have this issue.
- K-fold validation and iterated K-fold validation can address this problem.
- Split data into `K` partitions of equal size.
- For each partition `i` train a model on the remaining `K-1` partitions and evaluate on partition `i`

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# Iterated K-Fold Validation with Shuffling

- When have very little data available and need to evaluate model as precisely as possible.
- Apply K-fold cross-validation multiple times
  - Helpful if can call a K-fold with random shuffle as a reusable function.
- Final score is the average of the K-fold iterations (where are averages over K trainings).
- Expensive, end up training $P \times K$ models where $P$ is number of K-fold iterations and $K$ is number of folds used each time.

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# Beating a Common-Sense Baseline

- The only feedback you have is your validation metrics.
- Particularly important to tell whether or not you are actually able to predict anything at all or not.
- Before you start working with a dataset, you should always pick a trivial baseline that you'll try to beat.
  - If you beat that threshold, you'll know your model is at least doing something.
- If you have a binary classification problem where 90% of samples belong to class A, a classifier that always picks A achieves 90% accuracy.
- For classification, picking at random is often a basic baseline to beat.
- For regression, always guessing the mean or median output value is a low baseline.

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# Things to Keep in Mind about Model Evaluation

- **Data representativeness** You want both your training set and test set to be representative of the data at hand. For this reason you usually should at least **randomly shuffle** your data before splitting it into training and test sets.
- **The arrow of time** On the other hand, f you are doing time series prediction, you should **NOT** randomly shuffle your data before splitting it, this will cause **temporal leak**. Instead always make sure all data in test set is *posterior* to the data in the training set.
- **Redundancy in your data** Be careful of repeated data points appearing multiple times in a dataset (not uncommon). If you split a repeated sample and end up with it in the train and test set, you'll be leaking information and essentially testing with part of the data you trained with.

Having a reliable way to evaluate the performance of your model is how you'll be able to monitor the tension at the heart of machine learning—between optimization and generalization, underfitting and overfitting.

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# L05.3 Improving Model Fit

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# L05.4 Improving Generalization

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# Dataset Curation

Generalization in a deep learning model originates from the latent structure of your data. (You are only as good as your data).

- Make sure you have enough data.
  - You need a **dense sampling** of the input-cross-output space.
- Minimize labeling errors
  - visualize your inputs and check for anomalies and outliers, proffread your labels.
- Clean your data and deal with missing values.
- Do feature selection, analyze correlation or importance of features in a decision tree.

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# Feature Engineering

- **Feature Engineering** is the process of using your own knowledge about the data to make the algorithms work easier
- The essence of feature engineering: making a problem easier by expressing it in a simple way.

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# Using Early Stopping

In deep learning we always have models that are vastly operparamiterized: they have way more degrees of freedom than the minimum necessary to fit to the laten manifold.

- **You never fully fit a deep learning model**.
- Finding the exact point during training where generalization (validation loss) starts to decrease is one of the most effective things you can do to improve generalization.

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# Regularizing your Model

Regularization techniques are a set of best practices that actively impede the model's ability to fit perfectly to the training data, with the goal of making the model perform better during validation.

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# Adding Dropout

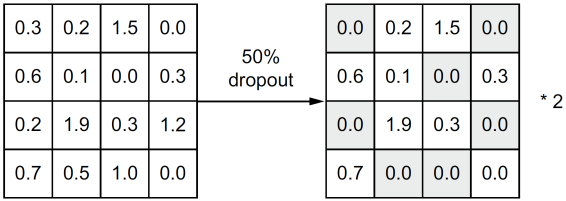


Figure 12: Dropout applied to an activation matrix at training time, with rescaling happening during training. At test time the activation matrix is unchanged. (Chollet, 2021, pg.150)

- **Dropout** is one of the most effective and most commonly used regularization techniques for neural networks.
- Dropout, applied to a layer, consists of randomly *dropping out* (setting to 0) a number of output features during training.
- The **dropout rate** is the fraction of the features that are zeroed out.
- At test time, we scale down output by the dropout rate.
- The core idea is that introducing noise in the output values of a layer can break up happenstance patterns that aren't significant.

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# Bibliography

Chollet, F. (2021). *Deep learning with python* (second). Manning.

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